

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

# RESONANCE26



## Round 1

ONLINE ROUND OF PPT SUBMISSION THROUGH PLATFORM



## ROUND 2

TOP SELECTED TEAMS PRESENT LIVE BEFORE JUDGES. ON 17TH OF APRIL.

## DOMAINS

- AI & ML
- CYBER SECURITY
- IOT
- OPEN INNOVATION

## PRIZE POOL

40,000+  
WINNER

## REGISTER NOW



**FACULTY MENTOR**  
Prof.Dr.Jyoti Kanjalkar

**HOD CSE(AIML)**  
Dr.Premanand Ghadekar

## 17<sup>TH</sup> APRIL



Venue :- Final offline round at VIT Pune Bibwewadi

| Event coordinators |

Chairperson :- Aman Shaikh (9607684669)

Vice-chair :- Tanvi Naik (9096576044)

# RESONANCE26

# RULEBOOK

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## SPONSORS:

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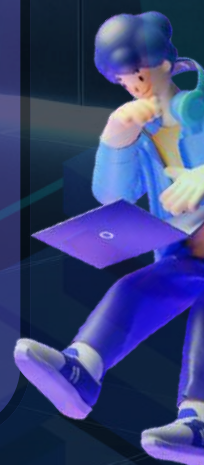
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- Overview
- Tentative Timeline
- Problem statements
- Submission



# OVERVIEW

**RESONANCE 2K26 is a fast-paced hackathon built for people who prefer building over just ideating.**

**In a limited timeframe, participants will work in teams on industry sponsored problem statements in AI/ML, Cybersecurity, IoT, and a bonus Open Innovation track.**

**Teams are expected to start early, develop real systems, and demonstrate clear progress through working prototypes or structured implementations.**

**If you can think clearly, build quickly, and iterate effectively, this is where you test it.**

# TENTATIVE TIMELINE

## Registration on Unstop

8<sup>th</sup> to 13<sup>th</sup> April 2026

Participants are expected to **submit** their **PPTs**, **Github** repos(optional), and **payment proof** upon registration

## PPT Submission Deadline

15<sup>th</sup> April 2026

Shortlisted teams will present their solutions and demos to the judges at the Grand Finale

## Grand finale at VIT Pune Campus

17<sup>th</sup> April

Shortlisted teams after screening round will be invited to VIT Pune campus for a final showdown.

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# PROBLEM STATEMENT

## 1) Sponsored Track: AIML in HealthCare

The AI/ML Track focuses on building intelligent systems that can perceive, analyze, and generate meaningful insights from real-world data.

This track emphasizes:

- Applied Machine Learning & Deep Learning — not just training models, but solving real problems
- Computer Vision, NLP, and Multimodal Systems working in production-like pipelines
- Reliability & Generalization — handling edge cases, diverse inputs, and real-world noise

Participants should aim to build systems that include:

- Data ingestion and preprocessing pipelines
- A well-justified modeling approach (CV/NLP/Hybrid)
- Structured outputs or visualizations that are interpretable
- A functional interface (web/mobile/dashboard)

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# PROBLEM STATEMENT

## AI-Powered Facial Skin Health Screening Tool

### ■ BACKGROUND

Access to dermatological care remains out of reach for millions due to high consultation costs, long wait times, and geographic barriers. Early and accurate detection of skin conditions is critical for effective treatment — yet most individuals have no immediate, accessible way to assess their skin health at home.

### ■ PROBLEM STATEMENT

Build an AI-driven web or mobile application that allows users to upload a clear photograph of their face and receive an automated, non-diagnostic skin health report. The tool must analyze the image across multiple dimensions and deliver structured, visual results with overlay annotations directly on the uploaded photograph, using computer vision and/or machine learning.

### ■ REQUIRED ANALYSIS MODULES

#### 01 Acne Severity

Grade from Clear → Mild → Moderate → Severe with a labeled scale indicator

#### 02 Lesion Count & Bounding Boxes

Detect 0–2, 3–5, 5–10, or 11–15+ lesions; draw color-coded bounding boxes on each detected lesion

#### 03 Facial Zone Mapping

Segment forehead, cheeks, nose, chin & jawline; highlight affected zones with outlined overlays

#### 04 Hyperpigmentation Overlay

Estimate dark spot coverage (0–10% to 30+%); outline each pigmented region directly on the face image

### ■ VISUAL OVERLAY REQUIREMENT

- ◆ **Face Mesh Grid:** Render a 3D landmark mesh over the face to show analysis coverage area
- ◆ **Lesion Bounding Boxes:** Color-coded boxes (green = comedonal, red = inflammatory, blue = other) with lesion IDs
- ◆ **Hyperpigmentation Outlines:** Traced region boundaries over dark spots showing distribution across the face
- ◆ **Zone Segmentation:** Outlined facial zones (forehead, cheeks, nose, chin/jawline) highlighting affected areas

### ■ PROGRESS TRACKING

Support multi-scan comparison across three timeframes — **Now** (current condition), **Short Term** (visible texture improvement, reduced breakouts), and **Long Term** (cleared breakouts, smoother and more even skin). Each stage should display the facial photo alongside annotated condition bullet points.

### ■ BONUS POINTS FOR

- ◆ Heatmap intensity overlay showing lesion density across facial zones
- ◆ Robust support for diverse skin tones and varied lighting conditions
- ◆ AI-generated skincare recommendations based on detected conditions

(as provided from sponsors verbatim)



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# PROBLEM STATEMENTS

## 2) Sponsored track: CyberSec Sponsored By : Passion InfoTech

### OVERVIEW

The Cybersecurity Track focuses on building **intelligent, automated security and governance systems** that can analyze, detect, and respond to real-world digital threats and compliance challenges.

Participants are expected to go beyond simple scripts or isolated models and instead develop end-to-end security pipelines, involving (not limited by):

- Data ingestion (logs, docs)
- Threat detection / validation logic (rules, ML, or hybrid systems)
- Explainable outputs



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# PROBLEM STATEMENTS

Participants can choose **ANY ONE** of problem statements below, all projects will be evaluated common rubrics.

## 1) Verify-ID (Visual Processing + Identity Governance)

### Focus:

Automating identity verification and fraud detection in digital onboarding systems.

### Expected Direction

- Extract structured data from government IDs (OCR-based)
- Perform face matching with live webcam input
- Validate ID formatting using rule-based checks

### Example Outputs

- Verification report with pass/fail reasoning
- Flags such as (Invalid DOB format / Face mismatch / Missing fields)

## 2) Log-Sentinel (Cybersecurity + Anomaly Detection)

### Focus

- Detecting suspicious activities from large-scale system logs.

### Expected Direction

- Parse and structure .log files
- Apply anomaly detection (e.g., Isolation Forest)
- Identify "impossible" or suspicious events

### Example Outputs

- Top 5 anomalies ranked by severity
- AI-generated security briefing

Example:

- "Multiple failed logins detected from foreign IP range"



### 3) Doc-Auditor (RPA + Compliance Automation)

#### Focus

Automating compliance checks for invoices, contracts, and financial documents.

#### Expected Direction

- Extract text from PDFs, Validate required fields (GST ID, totals, etc)
- Apply rule-based or LLM-based validation

#### Example Outputs

- Compliance status report per document
- Auto-generated correction email for missing fields

### 4) Phish-Guard (Cybersecurity + Email Analysis)

#### Focus

• Detecting phishing attempts using technical and visual deception patterns.

#### Expected Direction

- Analyze raw email headers, validate SPF and sender authenticity
- Detect look-alike domains using string similarity

#### Example Outputs

• Risk scorecard/ Breakdown of detected phishing techniques/  
Domain spoofing/ Header inconsistencies/ Unicode deception

### 5) Sovereign-Code-Audit (Software Governance + OSS Compliance)

#### Focus

- Ensuring SE projects comply with licensing and governance policies.

#### Expected Direction

- Analyze dependencies and LICENSE files
- Compare against a governance dataset (JSON-based)
- Flag risky or non-compliant dependencies

#### Example Outputs

- Compliance report
- Suggested safer alternatives for flagged libraries

## What does the Cybersecurity Track expect?

Every submission must demonstrate:

### 1. End-to-End Security Pipeline:

- Input → Analysis → Decision → Output
- Ability to clearly explain system flow
- Handling of real-world data formats (logs, PDFs, images, etc.)

### 2. Threat Detection / Validation Layer

- Rule-based logic, ML models, or hybrid approaches
- Clear reasoning behind detection decisions
- Ability to identify anomalies, inconsistencies, or risks

### 3. Explainability (Critical Requirement)

- System must justify its outputs
- Human-readable reports (NOT just graphs or flags)
- Example:  
"Login attempt flagged due to unusual IP + abnormal time pattern"

### 4. User-Facing Output

- Dashboard, report generator, or alert system
- Clear prioritization (e.g., risk levels, top threats)
- Actionable insights (not just detection)

### 5. Practical Implementation

- Should work on realistic datasets or simulations
- Lightweight and efficient solutions preferred
- No need for heavy training pipelines, focus on logic and system design

### Final Note

The goal of this track is to build practical, explainable cybersecurity systems, not just detection models.

The winning project will be the one that best combines:

- Strong system design
- Clear and defensible reasoning
- Real-world applicability
- Effective and usable outputs

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# PROBLEM STATEMENTS

## 3) Sponsored track: Internet of Things Sponsored By: DesignByte

The IoT Track focuses on building **end-to-end intelligent systems** that connect the **physical and digital world**.

Participants are encouraged to go beyond isolated components and develop complete, functional pipelines involving (not limited by):

- Sensor data acquisition pipelines
- Processing and decision-making (your intelligence layer)
- User-facing output (dashboard, alerts, automation)

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# PROBLEM STATEMENTS

Participants can choose **ANY ONE** of problem statements below, all projects will be evaluated common rubrics.

## 1) Smart Agriculture & Food Systems

### Focus

New innovations in Agri Tech, or systems that improve agricultural efficiency and productivity.

### Expected Direction

- Build solutions that use real-time data (sensors, imaging, etc.) to enable better decision-making in farming
- Focus on automation or predictive insights that reduce manual effort and resource wastage

### Example Projects (don't copy these)

- Greenhouse automation (temp/humidity control)
- Cold storage monitoring with spoilage alerts

## 2) Sustainable Infrastructure & Climate IoT

### Focus

- Optimizing resource usage and improving sustainability in urban and industrial environments.

### Expected Direction

- Develop systems that monitor and optimize energy, water, or waste usage
- Incorporate ML/AI, data visualization and actionable alerts for better infrastructure management

### Example Projects (don't copy these)

- Smart energy monitoring and optimization system
- Air quality tracking with visualization dashboards

## Design Bytes

### 3) Assistive Tech & Healthcare IoT

#### Focus

Enhancing accessibility, safety, and healthcare outcomes using connected devices.

#### Expected Direction

- Build solutions that provide real-time monitoring and alerts for critical conditions
- Focus on usability, reliability, and impact on quality of life

#### Example Projects (don't copy these).

- Wearable health monitoring with alerts
- Fall detection system for elderly care
- Navigation aid for visually impaired users
- Remote patient monitoring dashboard

### 4) Consumer IoT & Smart Experiences

#### Focus

Interactive, user-centric IoT systems designed to improve everyday experiences.

#### Expected Direction

- Create systems that emphasize user interaction, convenience, and personalization
- Integrate multiple devices/services into a seamless smart experience

#### Example Projects (don't copy these).

- Smart home automation (lighting, appliances)
- Smart mirror with personalized data display
- IoT-based gaming or gesture-controlled systems
- Productivity-enhancing connected devices

## What is Considered a Good Project?

A strong IoT project typically demonstrates:

### **1) Clarity of Problem:**

- Can you justify why IoT is needed for your defined problem?
- Clearly defined real-world use case
- Justified IoT use

### **2) Working Prototype: How close for real world deployment can you build it?**

- Working system
- Real-time or near real-time behaviour

### **3) Intelligence: Try to move beyond simple data visualization dashboards**

- Meaningful decision-making (alerts, automation, predictions)
- If AI/ML algos used, justify their exact purpose as well.

### **4) Solid Architecture: How well can it scale and how fragile is it?**

- Well-defined components (edge, cloud, communication)
- Scalable and logically structured

### **5) Usability: Friction for user adoption**

- Clean dashboard or interface
- Easy to interpret outputs

## **Final Note**

The goal of this track is to build complete IoT systems.

The winning project will be the one that best combines:

- Strong system design
- Quantifiable real-world impact
- Effective implementation

# PROBLEM STATEMENTS

## Bonus track: Open Innovation

The Open Innovation Track focuses on building practical, product-driven solutions that solve real-world problems through thoughtful system design and usability.

Just like the domain-specific tracks, participants are encouraged to innovate across industries while maintaining clarity of purpose and execution.

Projects should ideally go beyond isolated features and demonstrate complete, usable systems, involving (not limited by):

- Problem identification and justification
- System design and workflow logic
- User-facing product (web app, mobile app, platform, tool)

# NO PRE-DEFINED PROBLEM STATEMENTS !

All Open-Innovation solutions must demonstrate:

## 1. Problem-Driven System Design

- Clearly defined real-world problem

## 2. Functional Product

- End-to-end working system (think beyond prototype screens)

## 3. Value Creation

- Tangible benefit to users (efficiency, accessibility, cost, etc.)

## 4. User Interaction

- Clear, interpretable outputs
- Intuitive, low friction UIs for new users.

## 5. Implementation Flexibility

- All tech stacks allowed
- You should be able to justify all major technical choices

All Open Innovation projects will be judged on the same criteria:

| Criterion                    | Description                                   | What we're looking for                                 |
|------------------------------|-----------------------------------------------|--------------------------------------------------------|
| <b>Problem Clarity</b>       | Understanding and framing of the problem      | How well-defined and relevant is the problem?          |
| <b>System Design</b>         | Architecture, workflow, and structure         | How clean and scalable is your system design?          |
| <b>Practical Feasibility</b> | Real-world usability and deployment potential | Can this realistically be used beyond the hackathon?   |
| <b>Innovation</b>            | Creativity and uniqueness of solution         | What differentiates your solution from existing ones?  |
| <b>User Experience</b>       | UI/UX quality and usability                   | How easy is it for users to interact with your system? |

## What is Considered a Good Project?

A strong Open Innovation project typically demonstrates:

### **1) Clarity of Problem**

- Clearly defined and relevant real-world problem
- Strong justification of why this solution matters

### **2) Functional Completeness**

- Working end-to-end system
- Demonstrable workflows and features

### **3) Product-oriented Thinking**

- Focus on usability and user needs
- Clear differentiation from existing solutions

### **4) System Design**

- Logical architecture and modular structure
- Scalable and maintainable design

### **5) Usability**

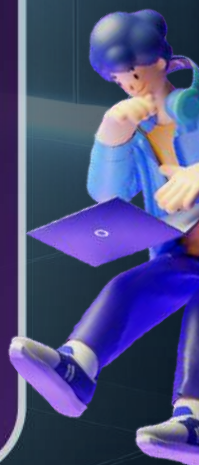
- Clean and intuitive interface
- Minimal friction for user adoption

## **Final Note**

The goal of this track is to build complete, usable products.

The winning project will be the one that best combines:

- Strong problem understanding
- Practical and scalable system design
- Clear real-world impact
- Effective implementation



# SUBMISSION & EVALUATION CRITERIA

## Submission Requirements

All teams are required to submit their work through the Unstop platform using the official PPT template provided in the attachments. Submissions that do not follow the prescribed format may be penalized during evaluation.

Participants must ensure they have the following requirements ready before submission:

- Payment Receipt and Proof
- Completed PPT (in PDF/PPT format) using the given template

Optional but Recommended :

- GitHub Repository Link (Should contain clean relevant project code and documentation)

## Important Constraints

- Adherence to PPT template is mandatory.
- Submissions must strictly follow the provided PPT format to ensure uniform and fair evaluation.
- GitHub Timeline Restriction : Any submitted repository **MUST NOT** contain commits **prior to 5th April**. This ensures fairness and that all teams build within the hackathon timeline.
- The use of AI tools (e.g., code generation, copilots, LLMs) is allowed, however, teams must be able to clearly explain their implementation, logic, and design decisions during evaluation.